Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Digital Logic

BCA — Semester I — 2021 — Answer Sheet

Subject Code: CACS103

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. Subtract: 453.35 - 321.17 using both 9's and 10's complement.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Simplify given Boolean function in both SOP and POS using k-map where d represents don't care condition.F (A, B, C, D) = ∏ (2, 3, 4, 5, 7, 9, 10) and ∏ d (0, 6, 11)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Define combinational logic circuit. Design a full adder with its truth table, logic diagram and Boolean expression.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define Decoder. Design BCD to Decimal decoder with its block diagram, truth table, circuit diagram and mathematical expression.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Explain basic gates and universal gate. Realize NOR gate as a universal gate.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain clocked JK flip flop with logic diagram, truth table, characteristic table and excitation table.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Differentiate between synchronous and asynchronous counter. Design 3-bit up ripple counter.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. Explain PLA with its block diagram. Design a combinational circuit using a ROM that accepts a 3-bit number and generates an output binary number equal to the square of the input number with circuit diagram, truth table and block diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is counter? Design a mod 7 synchronous counter with state table, state diagram, circuit diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define multiplexer. Explain 4:1 multiplexer with its block diagram, truth table and logic diagram. Mention how a 4:1 multiplexer works like lower order multiplexer.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
